

Graph using adjacency list: $\rightarrow$



Adj List: $\rightarrow$

0 $\rightarrow$	
1 $\rightarrow$	2, 3, 4
2 $\rightarrow$	1, 4, 5
3 $\rightarrow$	1, 4
4 $\rightarrow$	1, 2, 3, 5
5 $\rightarrow$	2, 4

TC = $O(2E)$

DU 2 $O(E)$

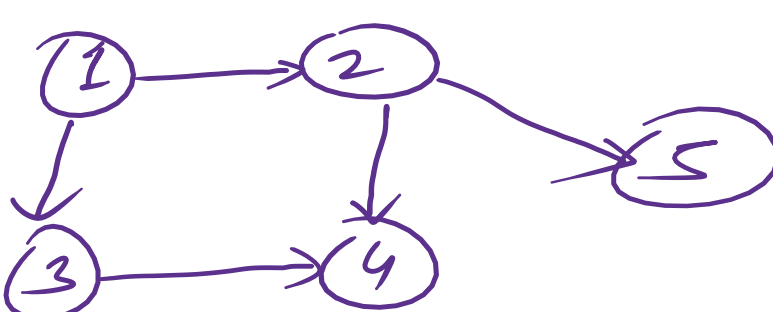
[i] $\rightarrow$ v

1 $\rightarrow$ 2, 3, 4

Node 1: 2, 3, 4

Directed Graph:

Adj-List $\rightarrow$



Adjacency List $\rightarrow$

0 $\rightarrow$

1 $\rightarrow$ 2, 3

2 $\rightarrow$ 4, 5

3 $\rightarrow$ 4

4 $\rightarrow$

5 $\rightarrow$

v.push(u);

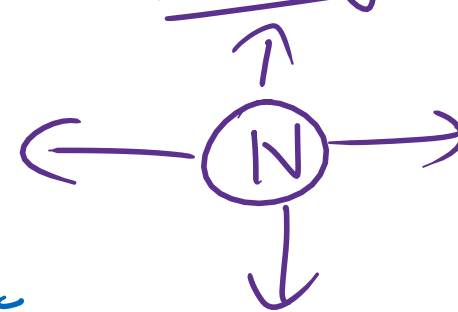
In degree

Out degree

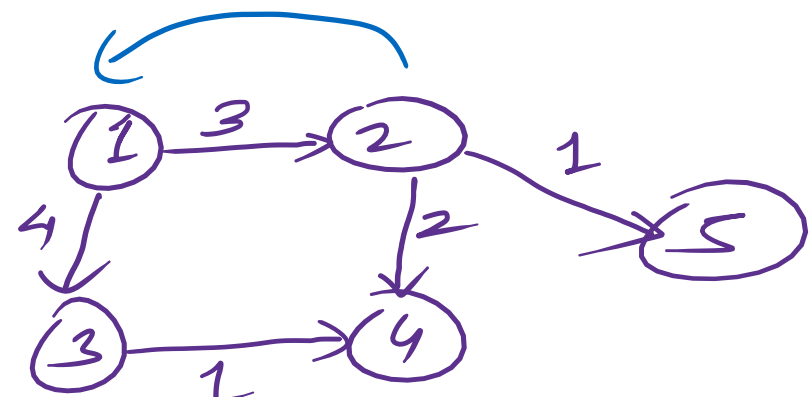


Towards

Aways



Weighted Directed Graph: $\rightarrow$



UD: Node: (1, 3)

5 5 N W N W

Node 1: (2, 3) (3, 4)

Node 2: (4, 2) (5, 1)

Node 3: (4, 1)

Node 4:

Node 5:

Undirected Weighted Graph AdjList H.W.

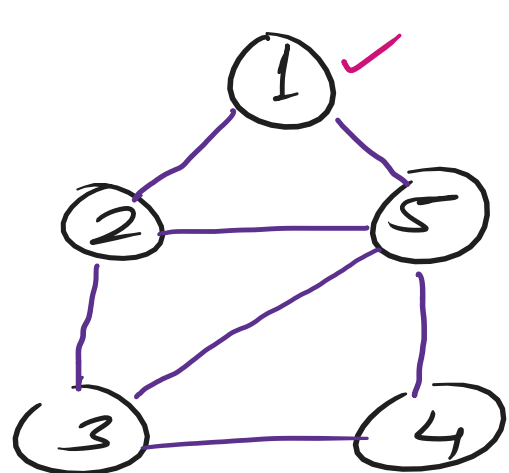
Graph Traversal Algorithms: $\rightarrow$ Adj List

① Breadth First Search

Level Order

Search Traversal

AM $\Rightarrow$ AL



Starting: ① $\rightarrow$ Who are your neighbours? [Adj List]

Adjacency List:

1 $\rightarrow$ 2, 5

$\Rightarrow$ 2 $\rightarrow$ 1, 3, 4, 5

3 $\rightarrow$ 2, 4, 5

4 $\rightarrow$ 3, 5

5 $\rightarrow$ 1, 2, 3, 4

Visited

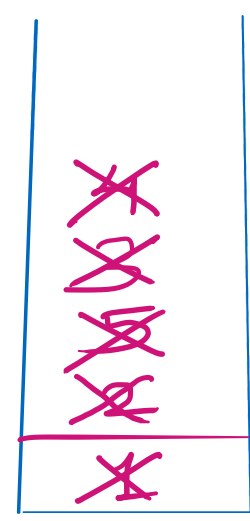
1 $\rightarrow$ T

2 $\rightarrow$ T

3 $\rightarrow$ T

4 $\rightarrow$ T

5 $\rightarrow$ T



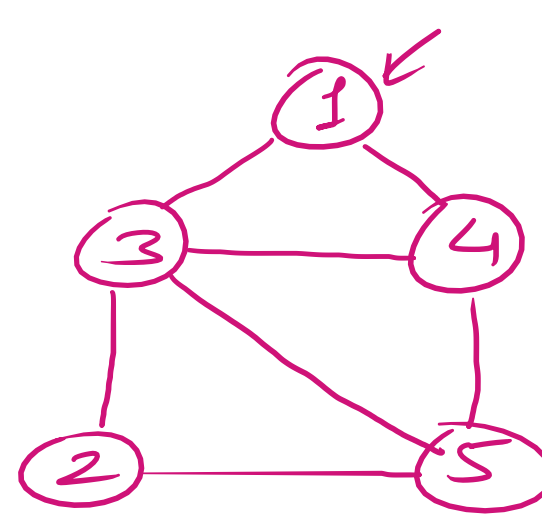
Output $\rightarrow$ 1, 2, 3, 4

queue

while (q.empty())

{

}



BFS $\rightarrow$ Traversal: $\rightarrow$

Starting Node $\rightarrow$ 1 w

BFS $\rightarrow$ any answer

1 2 4 3 1 $\rightarrow$ 1 3 4 2 5

Adjacency List:

it 0 0

1 $\rightarrow$ 3, 4

$\rightarrow$ 2 $\rightarrow$ 3, 5

$\rightarrow$ 3 $\rightarrow$ 1, 2, 4, 5

$\rightarrow$ 4 $\rightarrow$ 1, 3, 5

$\rightarrow$ 5 $\rightarrow$ 2, 3, 4

node

O/p: $\rightarrow$ 1, 3, 4, 2, 5

Visited

1 $\rightarrow$ T

2 $\rightarrow$ T

3 $\rightarrow$ T

4 $\rightarrow$ T

5 $\rightarrow$ T



queue